

Leonardo Santos

Electrical Engineer | Data Scientist | Software Developer

Electrical engineer with experience in electrical projects, integrated circuits, firmware, and software. Specialized in digital systems for signal processing and data science, with a focus on innovation and efficiency.

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🌐 je-leo-as.github.io

EDUCATION

Bachelor's Degree in Electrical Engineering

Federal University of Paraná

03/2019 - 12/2024

Courses

- Teaching assistant in Digital Electronics and Microelectronics, assisting with exercises, inquiries, and lab practices.
- Conducted a Python course for Chemical Engineering.
- Administrative manager at Seatel, organizing courses, lectures, and technical visits.
- Participant in PET: IoPET (IoT and firmware), BigPET (machine learning, leader), PET3D (design and 3D printing, leader).
- Developed electrical subsystem for an off-road vehicle (wiring harness, PCB, firmware, 3D modeling in Catia). Achievements: 1st place in electrical project Regional South 2019, 11th place national Baja SAE 2020, 2nd place in electrical project Regional South 2020.

Master's Degree in Electrical Engineering

Federal University of Paraná

03/2025 - Present

Courses

- Research in digital circuits for telecommunications, focused on signal processing and hardware optimization.
- Study of FPGA and ASIC architectures for energy efficiency in embedded wireless communication systems.

PROFESSIONAL EXPERIENCE

Software Developer / Data Scientist

IoTag Technology

10/2022 - Present

Company focused on precision agriculture

Achievements/Tasks

- Developed firmware in C/C++ for virtual terminal interfaces of agricultural machines.
- Created data science algorithms for spatial data and equipment collection.
- Implemented scalable cloud services using AWS and databases.

Engineering Intern

Lactec

06/2022 - 10/2022

Technology and research institute

Achievements/Tasks

- Assisted in the development of capacitive sensors for water level and flow for energy generation.
- Improved skills in electronics, microcontroller programming, and system integration.

ADDITIONAL ACTIVITIES

Member of Electrical Subsystem and Management

UFPR Baja SAE

01/2019 - 12/2021

Extension project for off-road vehicle

Achievements/Tasks

- Developed wiring harness, PCBs, and firmware.
- 3D modeling in Catia.
- Achievements: 1st place in electrical project Regional South 2019, 11th place national 2020, 2nd place Regional South 2020.

Research Scientist

GICS

08/2021 - 12/2024

Research group in circuits and systems

Achievements/Tasks

- Developed TRNG in VHDL, implemented in FPGA and designed with Cadence.
- Modeled DPD based on Memory Polynomial in Python, implemented in VHDL and validated with Xilinx.

Project Manager and Member

Tutorial Education Program (PET)

08/2021 - 12/2024

Undergraduate enhancement program

Achievements/Tasks

- IoPET: Developed an electronic lock with ESP01 and MQTT.
- BigPET: Led machine learning projects, taught a course, and developed a fake news detector.
- PET3D: Led design and 3D printing, including supports for face shield masks during the pandemic.
- Teaching assistant in Digital Electronics and Microelectronics.

PROJECTS

Virtual Assistant

- Multimodal assistant with LangChain for NLP, FastAPI for backend and speech recognition. Frontend in JavaScript.

Electronic Lock

- Smart lock with ESP-01 (ESP8266) and MQTT protocol for remote control.

TRNG Integrated Circuit

- TRNG in 130 nm with VHDL, optimized in FPGA, synthesized with Genus and routed with Innovus (43 cells, 41 μW).

Fake News Detector

- System for verifying news authenticity with text processing and machine learning (Logistic Regression, KNN, Random Forest).

Automatic Resume Translator

- Automated resume translation system with CrewAI and Typst, hosted on GitHub (Je-Leo-AS/Curriculo), with versions in Portuguese (PT-BR) and English (EN) in separate branches. Implemented GitHub Actions to trigger automatic translations via CrewAI, updating the EN branch.

n8n Automation — AI agents for maintenance and management of communications

- Automation platform with n8n to orchestrate flows and internal AI agents.
- Agent “Proxmox Care”: monitors the Proxmox server in the private network and performs maintenance tasks (resource checking, anomaly notifications, scheduling updates, backup routines, and cleaning). Works in an isolated environment with key authentication and restricted permissions to maintain network security.
- Agent “Comms Manager”: manages messages and emails from the professional number and email — ingests, prioritizes, summarizes, provides standard replies, and routes to channels (tickets, Slack, CRM). Uses NLP to detect urgency and extract relevant entities.
- Integration via webhooks, queues, and audit logs; automations triggered by events, with logs and notifications to ensure traceability and human review when needed.

SKILLS

Python

VHDL

C/C++

Java

JavaScript

PHP

HTML/CSS

3D Modeling (SolidWorks, Fusion 360, Catia)

Circuit Design (Cadence, ADS)

Circuit Simulation (Proteus)

PCB (Altium Designer)

Python Frameworks (OpenCV, FastAPI, LangChain)

Databases and SQL

Web Development

Interpersonal Communication

Planning

Time Management

N8N

CERTIFICATES

Android Development

60 hours

Machine Learning

60 hours

Databases

60 hours

LANGUAGES

Portuguese (native)

English (C1)